

ARGONYX'26

ROUND 2

Project Title: GuruGraph

Team Name: Team X

Team ID: T28

Team Lead: Samartha Puthraya K

One-line Tagline / Mission: Marks middle-school fraction homework from a phone photo, circles the exact wrong step, and gives the teacher tomorrow's 5-minute re-teach.

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ARGONYX'26

PROPOSED SOLUTION

THE PROBLEM

29%

average score on Class 6 fraction questions, their weakest maths skill

NCERT PARAKH Rashtriya Sarvekshan 2024

1 in 4

teachers say they have too much correction work

TISS SoTTTER 2023 · 3,615 teachers

A red cross says wrong. Not why.

HOW GURUGRAPH WORKS

Teachers don't upload anything.

1 Snap the page

A child snaps homework at home, or the teacher flips notebooks under a phone.

2 Find the exact wrong step

AI reads each line. Exact arithmetic finds the first wrong one and names the mistake.

3 Fix it by tomorrow

A short lesson in the child's language, and a 5-minute re-teach plan for the teacher.

GuruGraph

Asha · Class 7B · Fractions

ನನ್ನ ಹೋಮ್‌ವರ್ಕ್ ಪರಿಶೀಲಿಸಿ

3 ಲೆಕ್ಕ · 2 ಸರಿಪಡಿಸಬೇಕು

q1) $3/5 \div 3/10 = 2 \checkmark$

$= 5/3 \times 3/10 = 1/2 \times$

ಎರಡನೆಯದರ ಬದಲು ಮೊದಲ
ಭಿನ್ನರಾಶಿಯನ್ನು ತಿರುಗಿಸಿದ್ದೀರಿ

$= 15/30 = 1/2 \times$

$= 1/2 = 1/2 \times$

ಸ್ವಲ್ಪ ತಪ್ಪಾಗಿದೆ — ಎರಡನೆಯದರ ಬದಲು ಮೊದಲ
ಭಿನ್ನರಾಶಿಯನ್ನು ತಿರುಗಿಸಿದ್ದೀರಿ. ಸರಿಯಾದ ಉತ್ತರ 2.

A real handwritten page, read live and marked in Kannada

ARGONYX'26

TECHNICAL IMPLEMENTATION

TECHNICAL APPROACH

- 1 Phone photo**
Any page, any phone. Photos are never stored.
- 2 Gemini 3 Flash reads**
Every line of working, and where it sits.
- 3 Exact arithmetic judges**
Right or wrong, the first wrong line, the named mistake.
- 4 Agents act, teacher decides**
Coach plans, Analyst vetoes, Curator teaches in 3 languages.

REPOSITORY & DEMO



Try it live
class code 7B



Source code
GitHub, public

► **Demo video:** github.com/Argonyx-26/T28-Team-X/releases

Live: gurugraph-web-215071922486.asia-south1.run.app

Code: github.com/Argonyx-26/T28-Team-X

275 backend tests and 8 browser tests gate every deploy

Built with

FastAPI

SQLite

Next.js 16

Google Cloud Run

Gemini on Vertex AI

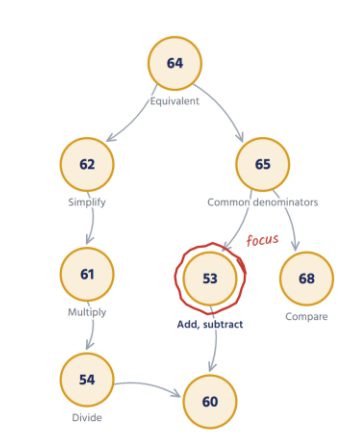
Chirp 3 HD voice

ARGONYX'26

PROTOTYPE & EXECUTION

What the class knows

class average per concept



Adding and subtracting fractions
Class average 53 (practising), 31 students assessed, 9 open gaps.
Most common: *added the denominators too* (14 students)

Who is stuck where

tap a student

	Equivalent	Simplify	Common denominators	Add, subtract	Compare	Multiply	Divide	Word problems
★ Asha	Green	Yellow	Yellow	Yellow	Grey	Red	Red	Grey
Aarav sim	Green	Yellow	Yellow	Green	Green	Yellow	Yellow	Green
Aishwarya sim	Green	Red	Green	Green	Green	Yellow	Yellow	Green
Ananya sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Arjun sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Bhavana sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Deepa sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Divya sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Diya sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Gowtham sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Harsha sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Isha sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Karthik sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
Kavya sim	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green
...	Green	Yellow	Green	Green	Green	Yellow	Yellow	Green

■ secure ■ practising ■ gap ■ not assessed yet

Teacher: who is stuck where

Problem 2: $2/5 \div 1/3$

X to fix

Q2) $2/5 \div 1/3 = 6/5 \checkmark$

$= 2/5 \times 1/3$ divided without flipping the second fraction $= 2/15 \times$

$= 2/15 = 2/15 \times$

Each line's exact value, in small type. The right answer is $6/5$.

Line 2: divided without flipping the second fraction. Not quite: you divided without flipping the second fraction. The answer is $6/5$.

Checked by exact arithmetic. Multiplying straight across without flipping the second fraction gives exactly $2/15$.

Every wrong line circled and named

A Analyst sent plan 1 back

No student shows 'added without making the denominators the same' on Adding and subtracting fractions; target a mistake the class actually makes.

C Coach revised the plan

Revised Re-teach group, 14 students Adding and subtracting fractions

Let's fix adding fractions: remember to not add the denominators too!

1. Display $2/3 + 1/3$ and ask students to solve it on mini whiteboards.
2. Look for common incorrect answers like $3/6$.
3. Ask a student who got $3/6$ to explain their thinking, focusing on 'added the denominators too'.
4. Guide the class to recall that only numerators are added when denominators are the same.
5. Work through the example $2/3 + 1/3 = 3/3 = 1$ together, emphasizing that the

Coach plans, Analyst vetoes

MEASURED ON REAL HANDWRITING

18/18

problems marked right or wrong correctly

13/13

wrong steps found

13/13

mistakes named

3.9 s

to read a whole page (median)

₹3.64

AI cost per student per month

Round 2: 6 phone photos of whole notebook pages by 3 writers, 18 problems (5 right, 13 with planted mistakes), labels written before the run. Cost: measured calls × published prices.

ARGONYX'26

BUSINESS MODEL

CUSTOMER

Middle-school maths teachers

Budget private schools first, then government schools through the state.

VALUE

Marked by morning

The wrong step named on every page, a lesson in the child's language, tomorrow's 5-minute plan.

MARKET

19,105 schools

private unaided schools in Karnataka. Only 8.2% of students there take coaching: the fix has to happen in school.

UDISE+ 2024-25 · MoSPI NSS Education 2025

BUSINESS MODEL

Per-student school licence

Free for a teacher's first class; the school pays to roll it out. State and district contracts for government schools.

₹3.64

AI cost per student per month, measured

SCALABILITY

One class → a district

Same unit everywhere: one child, one problem, one wrong step. Stateless API on Cloud Run; a new topic is a new rule pack.

₹5.1

to serve a student per month at 100,000 students (designed, list prices)

IMPACT

Every wrong answer becomes a named mistake, and a fix by the next morning.

LIVE TODAY

- ✓ Homework check at home
- ✓ Scan and snap mode
- ✓ 3 languages, with voice
- ✓ Coach vs Analyst plan debate
- ✓ School view

NEXT

- Classroom pilot: measure teacher time saved
- All middle-school maths topics
- WhatsApp homework
- District view for SCERT

No users or trials yet. Class 7B's 30 practice students are simulated, and labelled that way.

REFERENCES

RESEARCH PAPERS

[NCERT PARAKH Rashtriya Sarvekshan 2024](#)

[Muralidharan & Sundararaman 2010: diagnosis alone didn't raise learning](#)

[Siegler & Pyke 2013: how children get fractions wrong](#)

MARKET RESEARCH

[UDISE+ 2024-25](#) · [MoSPI NSS Education 2025](#) · [TISS SoTTTER 2023](#)

EXISTING SOLUTIONS STUDIED

[Mindspark](#) · [Khanmigo](#) · [Shiksha Copilot](#) · [Gradescope](#) · [Photomath](#)

APIS / DOCUMENTATION

[Gemini on Vertex AI](#) · [Chirp 3 HD voices](#) · [GuruGraph API docs](#)

DATASETS

[Our 9 handwritten pages, labelled before any model ran](#)

[Every evaluation with its n and method: /judges](#)